# Sahaj Bhadja

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## Summary

Software Engineer with 2+ years shipping production applications, concurrent services, real-time interfaces, and ML pipelines using Python, C++, TypeScript, React, PyTorch, REST APIs, WebSockets, distributed systems, automated testing, and performance optimization.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Operating Systems, Database Systems, Advanced NLP

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Data Structures, Object-Oriented Programming, Software Engineering, Computer Networks

## Experience

**USC Dornsife** **September 2025 – Present**
*Software Engineer* *Los Angeles, California*
- Restructured fault-tolerant asynchronous orchestration with explicit job states and failure isolation, processing 1K+ tasks while tripling throughput under concurrent workloads and preventing cascading failures across dependent services.
- Architected multi-tenant application infrastructure for 500+ users, reducing indexed-query latency from 1.2 seconds to 350 milliseconds through schema design, indexing, access-pattern optimization, and production performance analysis workflows.

**Grad8** **June 2025 – August 2025**
*Software Engineer Intern* *Los Angeles, California*
- Built React interfaces and Python services for uploads, grading history, configuration, and live processing feedback, reducing user turnaround 60% while supporting production workflows used by 150+ instructors.
- Designed concurrent multimodal processing with multithreading and secure ingestion, evaluating 10K+ text and handwritten submissions daily through recoverable execution paths while maintaining 90%+ grading consistency in production.

**USC AD Labs** **January 2025 – May 2025**
*ML Research Assistant* *Los Angeles, California*
- Built a PyTorch simulation pipeline processing 100K+ trajectory logs, improving long-horizon planning 18% through reproducible feature transformations, benchmark automation, regression testing, and detailed behavioral failure analysis workflows.
- Developed a spatial-temporal encoder combining map geometry, historical trajectories, and agent context, improving closed-loop stability 15% across complex scenarios through controlled evaluation and systematic root-cause debugging workflows.

**Aasma Technology Solutions** **August 2023 – July 2024**
*Software Development Engineer, SecureSync* *India*
- Engineered cross-platform microservices with automated VM builds and 20+ unit and integration suites using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS production release environments reliably.
- Optimized object-oriented API integrations and AWS data flows, improving response times 30% and pipeline efficiency 68% while reducing technical debt 20% through reusable services and structured reviews.

**Aasma Technology Solutions** **May 2023 – August 2023**
*Full Stack Developer Intern, VibrantChem* *India*
- Implemented eight microservices spanning authentication, permissions, billing, resources, customers, leads, and notifications, reliably synchronizing 2M+ records through REST APIs and event-driven WebSocket updates across distributed production workflows.
- Built responsive React workflows and automated Docker deployments through Jenkins CI/CD, achieving 95%+ successful validation and cutting release time 50% through repeatable configurations, staged testing, and monitoring.

## Skills

**Languages:** Python, C++, TypeScript, JavaScript, Java, SQL, Bash
**Applications:** React, Next.js, FastAPI, Flask, Node.js, REST APIs, WebSockets, Microservices, Responsive UI, OOP
**ML & Data:** PyTorch, TensorFlow, Transformers, Feature Engineering, Model Evaluation, PostgreSQL, SQL Server, MongoDB
**Systems & Quality:** Operating Systems, Concurrency, Distributed Systems, Linux, Networking, Unit Testing, Integration Testing, Debugging, Monitoring, Performance Optimization

## Projects

**CPU Scheduling Simulator** Live | *C++, React, Operating Systems*
- Implemented six preemptive and non-preemptive scheduling algorithms with React controls, real-time Gantt visualization, and dynamic latency metrics across 1K+ process combinations, applying systems fundamentals to interactive resource management.

**CaptionCraft - Real-Time ML Application** Live | *PyTorch, Transformers, Flask*
- Benchmarked BLIP, GIT, and ViT-GPT-2 across 10K images and 100K captions, improving measured accuracy from 81% to 96% while building scalable Flask inference APIs for responsive generation.

**FinNexus** Live | *TypeScript, Next.js, WebSockets, Kubernetes*
- Built a TypeScript application supporting 50K+ users and 1M+ transactions with responsive interfaces, secure integrations, real-time synchronization, advanced filtering, and reliable containerized deployment workflows at production scale.